

DOUGLAS MENDES BARRETO

Staff Software Engineer

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Professional Summary

Staff-level Software Engineer with **14+ years** of experience delivering software across mobile, web, backend, and platform engineering. My work centers on technical strategy, system design, modernization, and execution in environments that need both strong architecture and pragmatic delivery. I bring a hands-on approach to solving complex product and platform problems while raising engineering standards across teams.

Staff Engineering Strengths

- Technical direction for complex, multi-domain products
 - System design for scalable backend, mobile, and platform systems
 - Driving modernization across architecture, tooling, and workflows
 - Leading high-ambiguity initiatives from concept to production
 - Improving engineering effectiveness through automation and platform investments
 - Cross-functional partnership with product, design, and operations
 - Raising code quality through technical guidance and review
 - Translating business goals into executable technical plans
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Tech Skills

- Programming Languages: Python, TypeScript/JavaScript, Go, C++, C#, Ruby, Swift, Objective-C
 - Distributed Systems: REST APIs, GraphQL, microservices, event-driven architectures
 - Product Engineering: Node.js, React, Next.js, React Native, native iOS, native Android
 - Platform and Cloud: Google Cloud, AWS, Docker, Kubernetes, Terraform, Ansible, CI/CD
 - Data Systems: MySQL, PostgreSQL, MongoDB, Redis
 - Engineering Practices: system design, modernization, testing strategy, trunk-based development, design patterns, SOLID
 - Productivity and Automation: GitHub Actions, Jenkins, developer tooling, AI-powered development workflows
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Professional Experience

Senior Software Engineering Manager / Staff-Level Technical Lead

Xpertsea - Predictive shrimp biomass control with AI delivering sustainable aquaculture solutions

51 - 100 employees

Quebec, Canada

Oct 2020 – Present

- Set technical direction for AI-enabled product development across predictive analytics, operational tooling, and computer vision systems
 - Operated as the hands-on technical lead for a 9-engineer team, driving architecture decisions and delivery execution
 - Built internal AI prototyping workflows that reduced time-to-validation for new product ideas
 - Reduced lead time from 13 weeks to 2 weeks by redesigning engineering workflows, tightening delivery loops, and improving technical foundations
 - Introduced trunk-based development and Kanban to increase throughput and reduce coordination overhead
 - Rebuilt CI/CD pipelines, cutting build and deployment time from 50 minutes to 7 minutes
 - Led migration from AWS to Google Cloud, lowering infrastructure costs by 20%
 - Improved platform scalability and developer effectiveness through architectural and tooling investments
 - Partnered with product and design to translate roadmap goals into technically executable plans
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Senior iOS Developer

Social.mom - Social Media for mothers focused on delivering a safe and supportive environment

11 - 20 employees

Oct 2019 – Oct 2020

- Led re-architecture of the iOS codebase for a high-traffic social platform, improving maintainability and reducing feature delivery time by 20%
 - Directed migration from Objective-C to Swift to modernize the native stack and improve long-term development velocity
 - Built automation tooling that eliminated hundreds of hours of repetitive operational work
 - Established unit and end-to-end testing practices that increased release confidence and engineering discipline
 - Served as the primary technical reference for iOS architecture and code quality
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Senior Software Engineer

Levio - Tech Consulting

1K - 5K employees

Jul 2018 – Oct 2019

- Engineered high-availability backend services for multiple client environments using .NET C#
 - Led modernization of legacy systems, migrating critical workflows from COBOL to modern .NET platforms
 - Improved engineering collaboration by driving source control migration from legacy systems to Git
 - Worked as a technical consultant with stakeholders and delivery teams to translate requirements into maintainable solutions
 - Strengthened delivery reliability through CI pipeline improvements and automation
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Senior Mobile Engineer

Concrete Solutions - Tech Consulting

10K+ employees

Jul 2015 – Aug 2018

- Expanded into senior-level mobile engineering ownership, leading architecture and development for native applications across financial and retail clients
 - Delivered 3 mobile banking applications and 4 e-commerce apps in regulated and high-scale environments
 - Architected a white-label e-commerce mobile platform that enabled rapid rollout of customized storefronts from a single codebase
 - Designed CI/CD pipelines with Jenkins, Bitrise, and Travis CI to improve release reliability and team throughput
 - Raised engineering quality through technical talks on coding standards, testing practices, and QA implementation
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Lead Game Programmer

Editora Globo - Publisher and developer of digital entertainment products

500 - 1K employees

Aug 2010 – Apr 2015

- Took on technical leadership for cross-platform game development across iOS, Android, and Windows using C# and Unity3D
 - Architected backend systems for multiplayer experiences, including login, room control, and persistent game services
 - Created reusable Unity3D framework layers and tooling that increased team productivity
 - Developed large-scale gameplay and platform systems for educational digital products
 - Improved team execution by introducing JIRA-based planning workflows
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Junior Analyst Programmer

People Way - Tech Consulting

11 - 50 employees

Apr 2009 – Feb 2010

- Developed enterprise web applications using J2EE, JSP, JSF, and SOAP web services
 - Designed MySQL-backed data workflows and reporting integrations with JasperReports
 - Built parsers and automation scripts for SMS, XML, and CSV data processing
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Education

University Anhembi Morumbi

Bachelor's Degree in Game Design and Development - 2010

Languages

- English: Fluent
- Portuguese: Native
- French: Professional

Main Projects

Kampi (Xpertsea)

Intelligent copilot for shrimp farmers using AI and computer vision to replace manual guesswork with data-driven insights. Features three pillars: Pond Vision for biometric monitoring via smartphone photos, Optima for predictive analytics and growth forecasting, and FarmBook for team coordination across the production chain.

Tech Stack: Python, TypeScript/JS, MongoDB, MySQL, Node.js, Bun, Next.js, Google Cloud, Terraform, Ansible, C++, Android, iOS, React Native + Native Development

Link: <https://bekampi.com/>

- Led technical strategy for an AI-powered platform combining computer vision, predictive analytics, and farm operations workflows
 - Defined architecture and delivery approach for predictive systems, mobile experiences, and computer vision capabilities
 - Guided development of an omnichannel product across iOS, iPadOS, and web
 - Drove infrastructure modernization through Google Cloud, Terraform, and Ansible
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Social.Mom (Social.mom)

Social network platform designed to counter social isolation during motherhood and reinforce the social safety net of families. Enables mothers to connect locally, plan activities, and access community groups and organizations in their area through a bilingual interface.

Tech Stack: iOS native, Android native, Java Spring Boot, React

Link: <https://social.mom/en>

- Led iOS architecture work for a high-traffic bilingual social platform focused on community and local connection
 - Modernized native foundations to reduce maintenance overhead and improve delivery speed
 - Introduced testing practices that strengthened reliability and release confidence
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Meu Carrefour (Carrefour Brazil)

All-in-one retail companion and cornerstone of Carrefour Brazil's omnichannel strategy, empowering millions of users to manage loyalty rewards, shop for groceries online, and streamline in-store visits through smart utility tools like barcode scanning and proximity services.

Tech Stack: iOS native, TypeScript, Node.js, Java

- Developed core mobile commerce capabilities, including barcode scanning and store discovery features
 - Integrated loyalty experiences with discount and cashback tracking
 - Built shopping list workflows with cross-platform synchronization
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Itaú Unibanco Mobile Banking (Itaú Unibanco)

Mobile banking application for one of Brazil's largest financial institutions, providing users with secure access to core banking services including account management and financial transactions.

Tech Stack: Windows Phone, iOS, .NET

- Developed secure authentication and account access capabilities for a large-scale mobile banking product
 - Built core account visibility features including balances and transaction history
 - Implemented transfers and payments within the core banking experience
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Mundo do Sítio (Editora Globo)

Persistent, multi-user virtual world designed to integrate pedagogical goals with immersive gameplay as a digital extension of the Sítio do Picapau Amarelo IP. Featured complex economy, social interactions, and library of educational mini-games for millions of Brazilian students.

Tech Stack: Unity 3D, Flash ActionScript (SWC), Java, SmartFox Server, HTML5/JS

- Designed full-stack systems to support high concurrency and low-latency multiplayer interactions
 - Built gameplay systems and real-time state synchronization for 30+ educational mini-games
 - Implemented authentication, profile registration, and parental control capabilities for a children's platform
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